

Total No. of Questions : 4]

SEAT No. :

P-5056

[Total No. of Pages : 2

[6187]-459

T.E. (IT) (Insem.)

MACHINE LEARNING

(2019 Pattern) (Semester - I) (314443)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

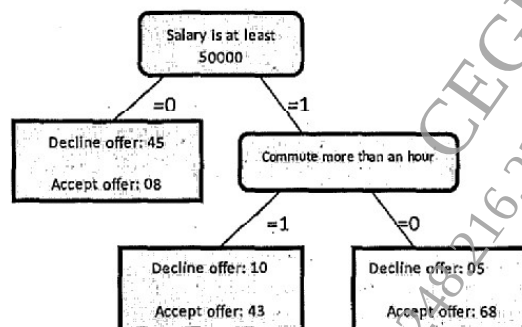
- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1) a) Compare supervised, unsupervised, and semi-supervised learning with examples (CO1). [6]
- b) Explain k-fold Cross Validation technique with example (CO1). [5]
- c) Why dataset splitting is required? State importance of each split in a machine learning model (CO1). [4]

OR

- Q2) a) Why size of training dataset is more compare to testing dataset? What should be ratio of Training & testing dataset? Explain any one dataset validation techniques. (CO1). [6]
- b) What is the need for dimensionality reduction? Explain the concept of the Curse of Dimensionality (CO1). [5]
- c) State and justify Real life applications of supervised and unsupervised learning. (CO1). [4]

- Q3) a) Consider the following feature tree. (Positive Class: Decline offer)



P.T.O.

Find :

[6]

- i) Contingency table
 - ii) Recall
 - iii) Precision
 - iv) Accuracy
 - v) False positive rate. (CO2)
- b) What is multiclass classification? Explain One-Vs-one construction method of multi class classifier with suitable example (CO2). [5]
- c) What are advantages and limitations of the Logistic Regression (CO2)? [4]

OR

Q4) a) Consider the following three class confusion matrix :

		Predicated			
		A	B	C	
Actual	A	14	3	3	20
	B	5	15	10	30
	C	2	5	43	50
		21	23	56	100

Calculate per class precision, per class recall, weighted average precision, weighted average recall and accuracy (CO2). [6]

- b) What is Support Vector Machine (SVM)? How does the SVM work (CO2)? [5]
- c) Why do we use Logistic Regression? Explain with suitable example (CO2). [4]

□□□

Total No. of Questions : 4]

SEAT No. :

P8579

[Total No. of Pages : 2

Oct-22/TE/Insem-559
T.E.(IT)
MACHINE LEARNING
(2019 Pattern) (Semester - I) (314443)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Show how machine learning differs from traditional programming. Elaborate with suitable diagram. [6]
- b) Explain K-fold Cross Validation technique with suitable example. [5]
- c) What is Dataset? Differentiate between Training dataset and Testing dataset. [4]

OR

- Q2)** a) Compare Supervised, Unsupervised and Semi-supervised Learning with examples. [6]
- b) What is the need of dimensionality reduction? Explain subset selection method. [5]
- c) What is feature? Explain types of feature selection technique. [4]

- Q3)** a) Consider the following three-class confusion matrix. Calculate Per-Class-Precision, Per-Class-Recall, weighted average precision, weighted average recall and accuracy. [6]

	Predicted Values			
		A	B	C
	A	45	10	05
	B	08	30	07
	C	06	04	40

P.T.O.

- b) Explain One-Vs-One construction method of multiclass classifier with suitable example. [5]
- c) Explain linear Support vector machine with suitable diagram. [4]

OR

- Q4)** a) What is multiclass classification? Explain One-Vs-Rest and One-vs-One multiclass classifier construction method with suitable example. [6]
- b) Write a short note on : [5]
- Various SVM kernel functions used to handle non-linear data.
- c) Define the following terms : [4]
- i) Accuracy.
 - ii) Precision.
 - iii) Recall.
 - iv) F1-score.

Total No. of Questions : 4]

SEAT No. :

PC-56

[Total No. of Pages : 2

[6360]-57

T.E. (IT) (Insem.)

MACHINE LEARNING

(2019 Pattern) (Semester - I) (314443)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Explain with example Predictive and Descriptive tasks of Machine Learning. Also state Predictive and Descriptive Model (CO1). [6]
- b) Explain k-fold Cros Validation technique with example (CO1). [5]
- c) Write a note on Principal Component Analysis (PCA) (CO1) [4]

OR

- Q2)** a) Justify which type of learning could be the most appropriate considering any one real world application of Machine Learning also explain your reasoning (CO1). [6]
- b) Explain Reinforcement Learning with diagram (CO1). [5]
- c) Discuss various scales of measurement of features in machine learning (CO1). [4]

- Q3)** a) What is multiclass classification? Explain One-Vs-Rest and One-vs-One multiclass classifier construction method with suitable example (CO2). [6]
- b) Explain true positive, true negative, false positive, false negative and class ratio for classification task (CO2). [5]
- c) Explain linear Support vector machine with suitable diagram (CO2). [4]

P.T.O.

OR

- Q4) a)** What is binary classification? Consider the following confusion matrix: [6]

		Predicated		Total
		+	-	
Actual	+	25	5	30
	-	10	20	30
Total		35	25	60

Calculate

- Recall
 - Precision
 - Accuracy
 - F1-score
 - Justify which evaluation measures will be more suitable for above matrix (CO2).
- b) What are support vectors in Support vector Machine? Explain Hard and soft margin SVM with the suitable diagram (CO2). [5]
- c) Write short note on Logistic regression (CO2). [4]

Total No. of Questions : 4]

SEAT No. :

PE55

[Total No. of Pages : 2

[6579]-356

T.E. (IT) (Insem)

MACHINE LEARNING

(2019 Pattern) (Semester - I) (314443)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.1 or Q.2 and Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain Types of Data, Scales of Measurement & Concept of Feature.[6]

b) Explain Leave one out cross validation LOO CV Technique with suitable example. [5]

c) Write a note on Principal Component Analysis (PCA). [4]

OR

Q2) a) Show how Machine Learning Differs from traditional Programming. Elaborate with suitable diagram. [6]

b) What is need of Feature selection Method? Explain different types of Feature selection Method. [5]

c) Explain Training and Testing dataset. Why size of Training dataset is more compare to testing dataset. [4]

P.T.O.

Q3) a) Consider the following 2 X 2 confusion matrix. **[6]**

Actual/Prediction	Covid-19 Positive (Prediction)	Covid-19 Negative (Prediction)	Total
Covid-19 Positive (Actual)	45	20	65
Covid-19 Negative (Actual)	05	30	35
TOTAL	50	50	100

Calculate its

- i) Precision
- ii) Recall
- iii) Accuracy
- iv) F1-score
- v) Justify

Which evaluation measures will be more suitable for above matrix.

- b) What is multiclass classification? Explain One-Vs-All construction method of multi class classifier with suitable example. **[5]**
- c) Explain Hyperplanes and Support Vectors. **[4]**

OR

Q4) a) Consider the following three class confusion matrix. **[6]**

		Predicated			
		A	B	C	
Actual	A	14	3	3	20
	B	5	15	10	30
	C	2	5	43	50
		21	23	56	100

Calculate per class precision, per class recall, weighted average precision, weighted average recall of accuracy.

- b) What is a role kernel in SVM? Explain various functions used to handle non-linear data. **[5]**
- c) Explain ROC (Receiver Operating Characteristics) curve. **[4]**

* * *